

Wearable device requirements

Project: ****

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Preface

This document was generated with the wearable device specification tool, which enables researchers to gather important requirements for a wearable device in a study, so that the device can be selected based on the requirements. In some situations, a device might not be able to meet all the requirements, and the researcher might need to make a decision on which requirements are most important. This document is a resource to guide the decision process between researchers and manufacturers.

Restore this configuration in the web-tool¹

General information

Type(s) of study: unspecified

Required devices: unspecified

Country of deployment/shipment: unspecified

Description of the intended main use case(s):

Hardware requirements

Housing & design

Aspect	Requirement	Device specs
Mounting	unspecified	
Housing material	unspecified	
Max dimensions (WxBxH)	5 x 5 x 2.5 cm	
Maximum weight (grams)	unspecified g	
Special design considerations	none	

¹Note that depending on the size of text inputs, restoring the bookmark may not work properly. The link will always contain all entered data, but url input length can be limited by the browser. To avoid silently dropped form-entries upon restoring the link, use a browser that supports long url inputs (e.g., Chrome) and be succinct with free-form text. Further, if the automatic redirect to the browser does not work when left-clicking on the link, copy the link via the right-click context menu and paste it manually into your browser of choice.

Measurement & sensors

Aspect	Requirement	Device specs
Min. covered spectral range (between 100 - 1000 nm)	380 - 780 nm	
Minimum number of channels across the spectral range	unspecified sensor channels	
Spectral sensitivity 1000nm	≥ No	
Min. operating range	1 - 10 ⁴ lx	
Field of view (sensors)	unspecified	
Modalities other than light	unspecified	
Sampling frequency	unspecified	

Battery & storage

Aspect	Requirement	Device specs
Battery: continuous recording runtime without recharge	≥2 weeks (with 10s recording interval & unspecified sampling)	
Storage: minimum on-device capacity	unspecified observations (≥unspecified days with 10s recording interval)	

Operating environment

Aspect	Requirement	Device specs
Operating temperature	-10 to 45 °C	
Operating humidity	20 – 95 % RH	
IP rating	unspecified	
Activity suitability	unspecified	

Controls & indicators

Aspect	Requirement	Device specs
Event marker button required	No	
Dedicated start/stop recording button	No	
User feedback indicators	unspecified	

Connectivity & I/O

Aspect	Requirement	Device specs
Acceptable connectors	unspecified	
Required wireless connectivity	unspecified	

Other hardware considerations

Data requirements

Software

Aspect	Requirement	Device specs
Operating systems (sync/export)	unspecified	
Software functions	unspecified	

Exported data

Aspect	Requirement	Device specs
Export format options	unspecified	
Required export variables	unspecified	
Timestamp format	unspecified	
Locale-independent export	Yes	
Store participant/session metadata	No	

Timestamps & recording interval

Aspect	Requirement	Device specs
Minimum recording interval	≤ 30 seconds	
Flexibility of recording interval	unspecified	
Exact recording intervals required	No	
Handle daylight savings transitions	No	
Retain correct date & time after battery drain	No	

Automated check and detections

Aspect	Requirement	Device specs
Automated detection & flagging	unspecified	

Storage

Aspect	Requirement	Device specs
Acceptable data storage locations	unspecified	

Other requirements

Budget

Aspect	Requirement	Device specs
Maximum Available budget per device, including shipping	unspecified	
Extended warranty/insurance required	No	
Acceptable device ownership options ²	unspecified	

Calibration

Aspect	Requirement	Device specs
Minimum required calibration validity	unspecified months	
Send recalibration information (cost, procedure, modalities)	No	

Miscellaneous

Aspect	Requirement	Device specs
Devices required (delivered) by	unspecified	
Free test device required	No	
Light-blocking, protective carrying case required	No	
Provision of easy-to-understand instructions suitable for study participants on correct device use.	No	

²**One-time purchase** means full ownership of hardware and software. **Rental model** means the devices are returned after an agreed upon time. Specification is required on how to handle broken or lost devices, and what falls under normal use. **Subscription model** means that the hardware is bought/owned, but that some software services - which need to be specified - are only available with an active subscription. This can include things like more in-depth analysis or cloud-storage.